

Low-Memory Neural Network Training: A Technical Report

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Abstract

Memory is increasingly often the bottleneck when training neural network models. Despite this, techniques to lower the overall memory requirements of training have been less widely studied compared to the extensive literature on reducing the memory requirements of inference. In this paper we study a fundamental question: *How much memory is actually needed to train a neural network?* To answer this question, we profile the overall memory usage of training on two representative deep learning benchmarks — the WideResNet model for image classification and the DynamicConv Transformer model for machine translation — and comprehensively evaluate four standard techniques for reducing the training memory requirements: (1) imposing sparsity on the model, (2) using low precision, (3) microbatching, and (4) gradient checkpointing. We explore how each of these techniques in isolation affects both the peak memory usage of training and the quality of the end model, and explore the memory, accuracy, and computation tradeoffs incurred when combining these techniques. Using appropriate combinations of these techniques, we show that it is possible to reduce the memory required to train a WideResNet-28-2 on CIFAR-10 by up to 60.7x with a 0.4% loss in accuracy, and reduce the memory required to train a DynamicConv model on IWSLT’14 German to English translation by up to 8.7x with a BLEU score drop of 0.15.

1 Introduction

Recently, there has been an explosion of interest around techniques to reduce the memory requirements of neural network *inference* (Han et al., 2016b, Zhu and Gupta, 2017). A central driver for this work is the fact that deep neural networks are notoriously parameter-hungry, typically containing several millions or even billions of parameters (Radford et al., 2018), making memory a significant hardware bottleneck for even storing and evaluating these models. Unfortunately, *training* is inherently more memory-intensive than inference (often by orders of magnitude), because it requires the storage of much more than just the network itself. However, the majority of state-of-the-art techniques to reduce inference memory are inapplicable to reducing training memory, as they often require first training a full-size model that is later compressed. As a result, training still requires